

(2.)

$$v_1 = (1 \ 2 \ 1)$$

$$v_2 = (0 \ 1 \ 2)$$

$$v_3 = (1 \ 0 \ -3)$$

$$v_4 = (1 \ 1 \ 0)$$

lahko t. vektor $\in \mathbb{R}^n$ zapisemo kot $\sum_{i=1}^m \text{lin. komb. } v_1, \dots, v_n, m \geq n$

lin. komb. je enolična, $m = n$

a) pokaži, da tvorijo ogradje $\mathbb{R}^3$; kateri 3je tvorijo bazo?

$$\begin{bmatrix} 1 & 0 & 1 & 1 \\ 2 & 1 & 0 & 1 \\ 1 & 2 & -3 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & -2 & -1 \\ 0 & 2 & -4 & -1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & -2 & -1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \Rightarrow \text{vektorji tvorijo ogradje,}$$

v_1, v_2, v_4 tvorijo bazo



lahko tudi v_1, v_3, v_4

ali: v_2, v_3, v_4

$$A(1 \ 2 \ 1) + B(0 \ 1 \ 2) + C(1 \ 1 \ 0) = (0 \ 2 \ 3)$$

$$A + C = 0 \rightarrow C = -A$$

$$2A + B + C = 2 \rightarrow 2A + \frac{3-A}{2} - A = 2$$

$$A + 2B = 3 \rightarrow B = \frac{3-A}{2} \quad 2A + 3 - A = 4$$

$$A = 1$$

$$C = -1$$

$$B = 1$$

$$\rightarrow (0 \ 2 \ 3) = v_1 - v_2 + v_4$$

③ v $\mathbb{R}^4$ so vektorji

$$v_1 = (1 \ 1 \ 0 \ 1)$$

$$v_2 = (2 \ 1 \ 1 \ 0)$$

$$v_3 = (1 \ 0 \ 2 \ 1)$$

Pokaži, da so $\downarrow$ lin. neodvisni in
jih dopolni do baze $\mathbb{R}^4$

$\downarrow$

$\{v_1, v_2, \dots, v_k\} \in \mathbb{R}^n$ so lin. neodv., če

ima enačba $\alpha_1 v_1 + \alpha_2 v_2 + \dots + \alpha_k v_k = 0$
samo eno rešitev in sicer $\alpha_1 = \alpha_2 = \dots = \alpha_k = 0$
 $k \leq n$

$$\alpha(1 \ 1 \ 0 \ 1) + \beta(2 \ 1 \ 1 \ 0) + \gamma(1 \ 0 \ 2 \ 1) = 0$$

$$\alpha + 2\beta + \gamma = 0$$

$$\alpha + \beta = 0 \rightarrow \beta = -\alpha = \gamma$$

$$\beta + 2\gamma = 0 \rightarrow 3\gamma = 0 \rightarrow \gamma = 0$$

$$\alpha + \gamma = 0 \rightarrow \alpha = -\gamma \quad \begin{matrix} \alpha = 0 \\ \beta = 0 \end{matrix} \Rightarrow \text{vekt. so lin. neodv.}$$

$$\left[\begin{array}{ccc|cccc} 1 & 2 & 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 2 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 & 1 \end{array} \right] \sim \left[\begin{array}{ccc|cccc} 1 & 2 & 1 & 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & -1 & 1 & 0 & 0 \\ 0 & 1 & 2 & 0 & 0 & 1 & 0 \\ 0 & -2 & 0 & -1 & 0 & 0 & 1 \end{array} \right] \sim \left[\begin{array}{ccc|cccc} 1 & 2 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 2 & 0 & 0 & 1 & 0 \\ 0 & 0 & 2 & -1 & 1 & 1 & 0 \\ 0 & 0 & 4 & -1 & 0 & 2 & 1 \end{array} \right] \sim \left[\begin{array}{ccc|cccc} 1 & 2 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 2 & 0 & 0 & 1 & 0 \\ 0 & 0 & 2 & -1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & -2 & 3 & 1 \end{array} \right]$$

$\Rightarrow$ lahko izberemo e_1, e_2, e_3 ali e_4 ; z vsemi bojo v_1, v_2, v_3 tvorili bazo

④ Poišči baze vekt. prost. 2×2 simetričnih in antisim. matrik

① $A \in S(2) = \{A \in \text{Mat}(2 \times 2, \mathbb{R}) \mid A = A^T\}$

$$A = \begin{bmatrix} a & c \\ c & b \end{bmatrix} \quad B = \left\{ \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}, \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \right\} \quad \dim S(2) = 3$$

② $\underline{so}(2) = \{W \in \mathbb{R}^{2 \times 2} \mid W = -W^T\}$

$$W = \begin{bmatrix} 0 & a \\ -a & 0 \end{bmatrix} \quad B' = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \quad \dim \underline{so}(2) = 1$$

$\dim S(n) = ?$

$$S(n) = \begin{bmatrix} a_1 & b_1 & c_1 & \dots \\ b_1 & a_2 & b_2 & \\ c_1 & b_2 & a_3 & \dots \\ \vdots & & & \ddots \\ & & & & a_n \end{bmatrix} \quad \begin{aligned} \text{št. param.} &: n + (n-1) + \dots + 1 = \\ &= \frac{n}{2}(n+1) \\ \Rightarrow \dim S(n) &= \frac{n}{2}(n+1) \end{aligned}$$

$\dim \underline{so}(n) = ?$

$$\underline{so}(n) = \begin{bmatrix} 0 & b_1 & c_1 & \\ -b_1 & 0 & b_2 & \\ -c_1 & b_2 & 0 & \ddots \\ \vdots & & & 0 \end{bmatrix} \quad \begin{aligned} \text{št. param.} &: (n-1) + (n-2) + \dots + 1 = \\ &= \frac{(n-1) \cdot n}{2} \\ \Rightarrow \dim \underline{so}(n) &= \frac{n(n-1)}{2} \end{aligned}$$

$$\dim S(n) + \dim \underline{so}(n) = \frac{n}{2}((n+1) + (n-1)) = \frac{n}{2}(2n) = n^2$$

$$\underline{\mathbb{R}^{n \times n} = S(n) \oplus \underline{so}(n)}$$

- Podobno tudi za hermitske ($H(n)$) in antihermitske ($\underline{u}(n)$)

- $\underline{so}(3) \rightarrow$ kotna hitrost tlesa v 3D



$$W = \begin{bmatrix} 0 & -w_z & w_y \\ w_z & 0 & -w_x \\ -w_y & w_x & 0 \end{bmatrix} \rightarrow W_x = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}$$

$$W_y = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{bmatrix}$$

$$W_z = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

IZOMORFIZM

$$\vec{\omega}: \mathfrak{so}(3) \rightarrow \mathbb{R}^3$$

$$(w_x W_x + w_y W_y + w_z W_z) \rightarrow (w_x, w_y, w_z) \text{ kotna hitrost}$$

⑤

$$V = \{ p \in \mathbb{R}_3[x] ; p(-1) = p(1) = 0 ; p'(0) = 0 \}$$

pokaži, da je V vekt. podprostor $\mathbb{R}_3[x]$, poišči kakšno bazo

$$\text{-stl. baza: } \{1, x, x^2, x^3\}$$

$$6) \quad U = \{(x, y, z, t) \in \mathbb{R}^4 / x - y + t = 0, y + z = 0\}$$

$$V = \text{span}\left\{ \underset{\substack{\text{"} \\ v_1}}{(0 \ 1 \ -1 \ 1)}, \underset{\substack{\text{"} \\ v_2}}{(1 \ 0 \ 0 \ 1)} \right\}$$

a) bazi $U \cap V, U + V$ $\dim U = 2$
 $\dim V = 2$

$$V \ni v = t_1 v_1 + t_2 v_2 = (0 \ t_1 \ -t_1 \ t_1) + (t_2 \ 0 \ 0 \ t_2) = (t_2 \ t_1 \ -t_1 \ t_1 + t_2)$$

$$\left. \begin{array}{l} x - y + t = 0 \rightarrow t_2 - t_1 + t_1 + t_2 = 0 \rightarrow t_2 = 0 \\ y + z = 0 \rightarrow t_1 - t_1 = 0 \rightarrow t_1 \in \mathbb{R} \end{array} \right\} U \cap V \text{ je generiran } (0, t_1, -t_1, t_1) \\ = z(0, 1, -1, 1)$$

$$\dim V + \dim U = 4 = \dim(U \cap V) + \dim(U + V) \rightarrow \dim(U + V) = 3$$

baza V:

z, t parametra $x = -z - t$

$$x - y + t = 0$$

$$y = -z$$

$$y + z = 0$$

$$V \ni u = \{t(-1, 0, 0, 1) + z(-1, -1, 1, 0)\}$$

$$U = \text{span}\left\{ \underset{\substack{\text{"} \\ u_1}}{(-1, 0, 0, 1)}, \underset{\substack{\text{"} \\ u_2}}{(-1, -1, 1, 0)} \right\}$$

baza $U + V$:

$$u_2 = u_1 - v_1$$

$$\Rightarrow \underline{U + V = \text{span}\{u_1, v_1, v_2\}}$$

b) poišči podpr. $W \subset \mathbb{R}^4$, da velja $(U + V) \oplus W = \mathbb{R}^4$

$$\dim W = 1$$

$$\left[\begin{array}{cccc|cccc} -1 & 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 & 0 & 1 & 0 \\ 1 & 1 & 1 & 0 & 0 & 0 & 0 & 1 \end{array} \right] \sim \left[\begin{array}{cccc|cccc} -1 & 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 2 & 1 & 0 & 0 & 0 & 0 \end{array} \right] \sim \left[\begin{array}{cccc|cccc} -1 & 0 & 1 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 1 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 \end{array} \right]$$

$\uparrow \uparrow$
 $W = \text{span}\{e_2\}$